nuclear isobars. As a result of this privileged character of even numbers, for every nucleus containing an even number of protons (having an even atomic number) there exist more or fewer stable isotopes—i.e. stable nuclei having the same atomic number and the same chemical properties, all of which are varieties of the same element. On the other hand, elements with an odd atomic number have far fewer stable isotopes. Thus, the element titanium, with its even number of protons (22), has as many as five stable isotopes, while its neighbour, vanadium, with its 23 protons, has only one single stable isotope. The next element, chromium, has again four isotopes, whereas the immediately following one, manganese, has just one. Cadmium, with its 48 protons, has as many as eight stable isotopes, whereas silver ($Z = 47$), which immediately precedes it, has only two. And so on throughout the entire periodic system.

The assumptions already discussed concerning the nature of nuclear forces, in particular their short range and their capacity for saturation, are thus definitely confirmed by actual experience in the form of conclusions based on them regarding the relative abundance and stability of the various nuclear species.

This disclosure of the existence of isotopes of the various elements furnishes an explanation of the fact that Prout’s hypothesis, based on the assumption that all the known atomic weights are integers—which was in some degree a movement in the right direction,—remained consigned to the limbo of oblivion for nearly a whole century. Subsequent measurements of the atomic weights of the heavier elements proved most of them to be very far from integers, or even approximately integers. But there is a simple explanation of this observation: the existence of isotopes. Every element made chemically pure is a mixture of isotopes (unless it has no more than isotope). The mass numbers of the individual isotopes are, in fact, always approximately integers. But chemical processes only give the average mass of the atoms in a mixture of isotopes depending on the proportions in which the isotopes are present, and which therefore may assume all possible non-integral values.